View-consistent 3D Venation Evidence Extraction from Transmittance-enhanced Multi-view Leaf Images

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1 Introduction

Three-dimensional plant phenotyping enables direct measurement of plant and organ geometry, but conventional 3D reconstruction mainly describes external surface structure and does not explicitly recover leaf venation. Venation is related to transport, mechanical support, development, and leaf morphology [3]. Most computational extraction approaches operate on detached, flattened, or scanned leaves [1, 2, 6]. Such 2D representations provide detailed vein information but lose the natural 3D pose and make repeated observation of the same intact leaf difficult.

We investigate a simple alternative based on transmittance-enhanced multi-view imaging. Backlighting improves the visibility of vascular structures while the leaf remains in its natural pose. Image-level leaf, boundary, and venation responses from calibrated views are projected onto a reconstructed surface and combined to obtain view-consistent 3D venation evidence. Rather than producing a complete biological vein graph, the proposed pipeline extracts a mesh-constrained venation candidate band on the natural leaf surface.

2 Methodology and Preliminary Results

An overview of the pipeline is shown in Fig. 1. Calibrated RGB images are acquired around an isolated leaf under backlighting and used for multi-view surface reconstruction [4, 5]. Leaf and venation evidence extracted from individual images is treated as view-level observation rather than final segmentation. The observations are projected onto the reconstructed mesh and combined across valid views, producing a 3D venation candidate band.

For preliminary quantitative verification, one pumpkin leaf was subsequently detached and imaged in a flattened state with a checkerboard reference. The flattened image is used only for evaluation and does not contribute to the 3D reconstruction. The reconstructed leaf is globally aligned with the reference for geometric comparison without applying non-rigid warping.

Using 14 calibrated views, the reconstructed boundary achieved a Chamfer distance of 1.136 mm to the flattened reference. The resulting 3D candidate band

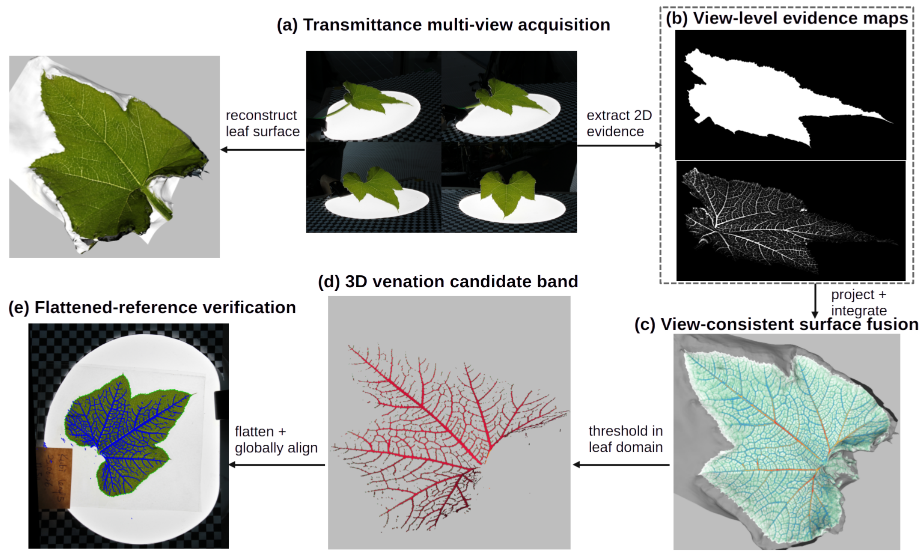

Fig. 1: Overview of the proposed pipeline and representative result. Transmittance-enhanced multi-view images are used to reconstruct the leaf surface and obtain image-level venation evidence. The evidence is combined on the reconstructed mesh to form a 3D venation candidate band, which is subsequently compared with an independently acquired flattened reference.

covered 96.1% of manually annotated key-vein points within a 1 mm tolerance. These measurements are intended as preliminary geometric verification of the candidate representation rather than evaluation of a complete venation graph. Additional qualitative tests on eggplant and cauliflower leaves produced visible surface-aligned venation evidence, while also indicating sensitivity to leaf optical properties and acquisition settings.

3 Conclusion

We presented a low-cost pipeline for recovering venation evidence on the natural 3D surface of a leaf using transmittance-enhanced multi-view images. Preliminary results demonstrate the feasibility of combining image evidence with reconstructed geometry to obtain a surface-constrained venation representation. The current study serves as a proof of concept, and broader quantitative evaluation across leaves, species, acquisition conditions, and alternative fusion strategies will be investigated in future work.

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